

Cloud Networking – Assignment 4

1. In AWS, create a new security group named InstaClone-SG and configure it to allow incoming HTTP (port 80) and HTTPS (port 443) traffic only from your own IP address.

Ans:

Step 1: Create the Security Group

A new security group named InstaClone-SG was created in AWS.

Step 2: Configure Inbound Rules

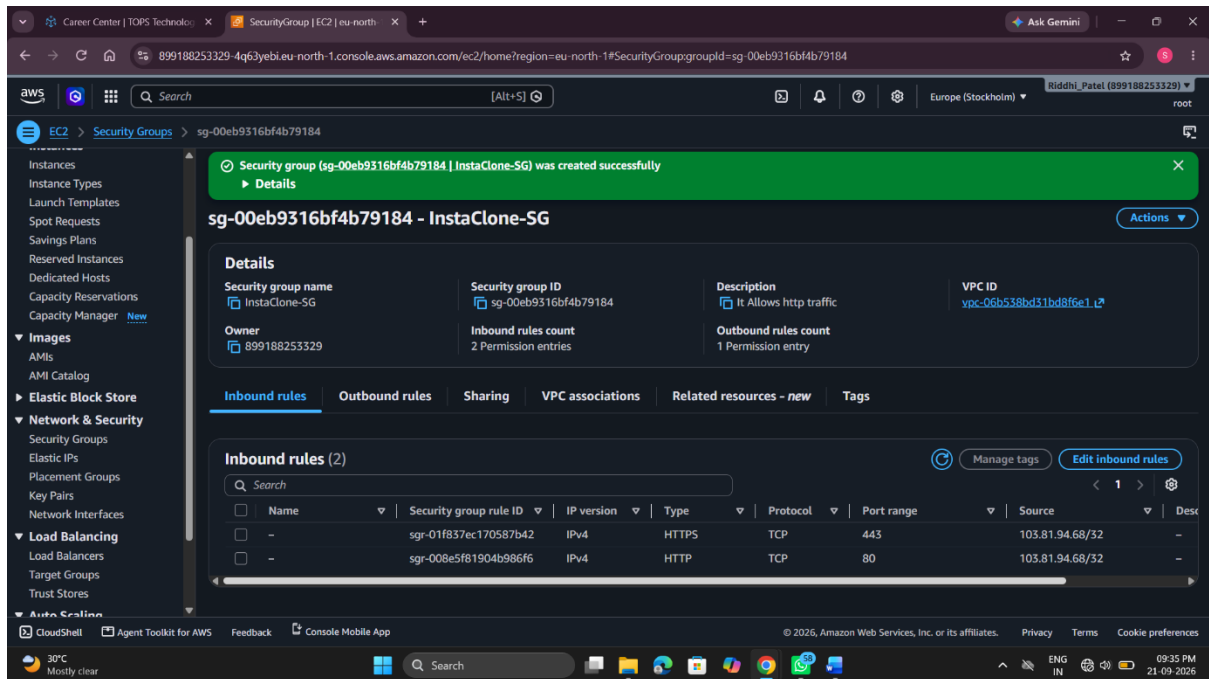
The following inbound rules were added:

- HTTP – Port 80 → Allowed only from My IP
- HTTPS – Port 443 → Allowed only from My IP

Step 3: Create the Security Group

The security group was created successfully with the configured inbound rules.

Screenshot 1: InstaClone-SG Inbound Rules – HTTP and HTTPS Allowed from My IP



2. Set up a security group for a mock Zomato-style backend server that allows incoming MySQL (port 3306) connections only from a specific IP address and blocks all other inbound traffic. Hint: Use the 'Custom TCP' rule and specify the allowed IP in CIDR format.

Ans: Step 1: Create the Security Group

Create a new security group for the mock Zomato backend server.

Security Group Name: Zomato-Backend-SG

Step 2: Configure the Inbound Rule

Add a Custom TCP inbound rule:

- Type: Custom TCP or MYSQL
- Port: 3306
- Source: Specific IP address in CIDR format, e.g. my ip/32

This allows MySQL connections only from the specified IP address.

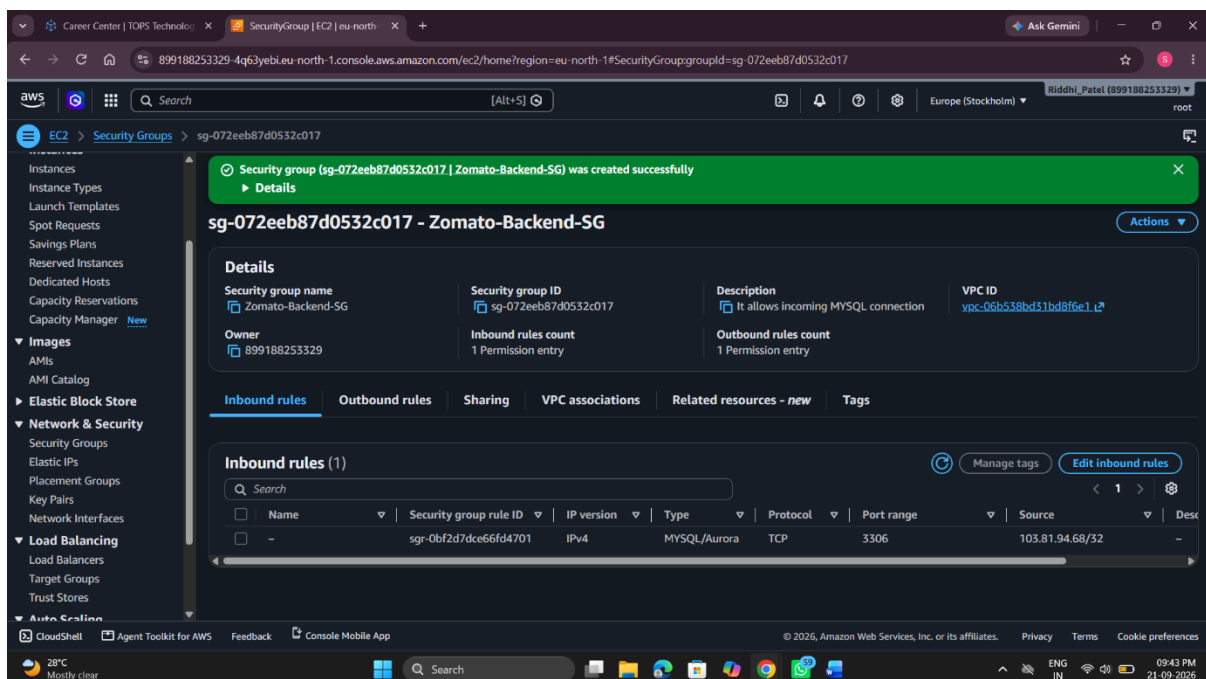
Step 3: Block Other Inbound Traffic

Remove any other inbound rules so that no other inbound traffic is allowed.

Step 4: Create the Security Group

Create the security group with the above configuration.

Screenshot 1: Zomato-Backend-SG – MySQL Port 3306 Allowed from Specific IP



Result:

The security group allows incoming MySQL traffic on port 3306 only from the specified IP address and blocks all other inbound traffic.

- 3. Modify an existing security group to block all outbound traffic except for DNS (port 53) and HTTPS (port 443) to simulate a strict Paytm wallet server environment.**

Ans:**Step 1: Modify the Existing Security Group**

The existing InstaClone-SG security group was selected and its outbound rules were modified.

Step 2: Remove the Default Outbound Rule

The default All traffic outbound rule was removed.

Step 3: Allow HTTPS Traffic

An outbound rule was added for:

- HTTPS – Port 443
- Destination: 0.0.0.0/0

Step 4: Allow DNS Traffic

An outbound rule was added for:

- DNS (UDP) – Port 53
- Destination: 0.0.0.0/0

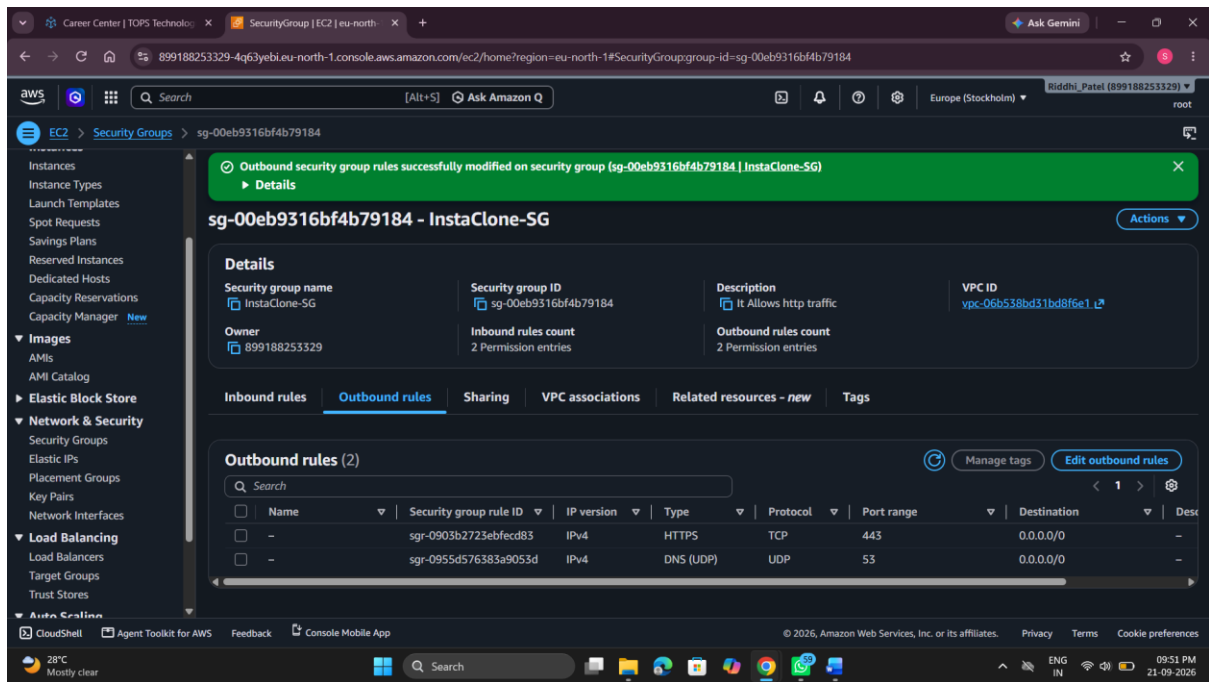
Step 5: Save the Rules

The changes were saved successfully.

Result

The security group now allows only **HTTPS (443)** and **DNS (53)** outbound traffic, while other outbound traffic is blocked.

Screenshot : InstaClone-SG Outbound Rules – HTTPS 443 and DNS 53 Only



4. Compare security groups and network ACLs by listing two key differences in how they control traffic for a Flipkart-like cloud application. Hint: Think about the level (instance vs subnet) and direction (inbound/outbound) each operates at.

Ans:

Security Groups	Network ACLs
1. They work at the instance level and control traffic to and from individual EC2 instances.	1. They work at the subnet level and control traffic entering or leaving the entire subnet.
2. They are stateful, so if inbound traffic is allowed, the response traffic is automatically allowed.	2. They are stateless, so inbound and outbound traffic must be allowed separately.

For a Flipkart-like application: Security Groups can protect individual web servers, while Network ACLs can provide an additional traffic-control layer for the entire subnet.